

Operations Research for Health Care

Lesson 12/15

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Outline of this lecture

Facility Location Under Uncertainty

Stochastic Programming

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Stochastic Programming

Uncertainty regarding the location of facilities

Facilities typically operate for years or **decades**. Costs, requirements and travel times *can change* significantly ¹. Decisions about the location of facilities are costly and **difficult to reverse**, and their effects extend over a long time horizon [Snyder, 2006].

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Parameter estimates may be inaccurate due to poor measurements or aggregation of demand points and choice of distance norm. We explore the two-stage nature of facility location problems (choosing locations now and assigning patients to facilities once uncertainty has been resolved).

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The **strategic** phase involves capital investment, followed by a **tactical** phase, with uncertainty being resolved in between. But firstly, the terms **certainty**, **risk** and **uncertainty** must be defined [Contreras and Ortiz-Astorquiza, 2019].

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Parameter-based decision type definitions

Certainty parameters are deterministic and known;

²The existence of a probabilistic description of uncertainty does not prevent the use of robustness measures.

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Risk Uncertain parameters with **known probability distributions**. Stochastic programming optimises the *expected* objective function. Parameters are probability distributions. *Identifying scenarios is difficult* (small numbers of scenarios for computations), but the advantage is that parameters are statistically dependent. E.g. demand vs time or locations; costs vs suppliers;

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Uncertainty Parameters are uncertain and **probabilities are unavailable**². This is often solved with robust optimisation, which optimises worst-case performance. If no probability information is known, *intervals* are assumed.

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Attitude towards risk and Decisions

The decision maker's **attitude to risk** are:

- ▶ **Risk neutral**: the decision maker does not consider risk when making a decision, and a linear function represents the associated return. If a probability can be assigned to each scenario, a risk-neutral decision maker will seek a decision that *minimises the expected cost (or maximises the expected return)*.

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- ▶ **Risk averse**: The decision maker wants to avoid unnecessary risk. The expected value of future assets is no longer an appropriate objective. Such a decision maker may seek to *minimise the maximum cost across all scenarios*.

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The **decisions** are:

- ▶ *ex ante*³ (**here and now**): decisions that need to be taken before uncertainty is revealed;

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The **decisions** are:

- ▶ *ex ante*³ (**here and now**): decisions that need to be taken before uncertainty is revealed;
- ▶ *ex post*⁴ (**wait and see**): decisions to be taken after the uncertainty has been revealed.

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Stochastic Programming

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If we know the joint probability distribution of the underlying random vector.

- ▶ Choosing an appropriate *performance measure* is part of the modelling process.
- ▶ **Stochastic location modelling**: first stage are location decisions; second stage are assignments of patients to facilities, i.e. recourse decisions. We generate equivalent problems.

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To illustrate, we work with some well-known problems. For a comprehensive overview of the foundations of stochastic programming, please refer to the seminal works of [Birge and Louveaux, 2011] and [Shapiro et al., 2021], and lecture notes of [Oliveira, 2024].

The **Uncapacitated** Facility Location Problem (UFLP)[Contreras and Ortiz-Astorquiza, 2019]

I : The set of all patients' origins.

J : The set of all (possible) destination health care locations.

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- ▶ D_i : Population requirement for health at origin $i \in I$ (pop).
- ▶ F_j : Setup cost for facility $j \in J$ (\$).
- ▶ C_{ij} : Unit cost for meeting demand (not full demand) from $i \in I$ to $j \in J$ (\$).

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- ▶ C_{ij} : Unit cost for meeting demand (not full demand) from $i \in I$ to $j \in J$ (\$).
- ▶ $x_{ij} \geq 0$: The **fraction** of demand of $i \in I$ supplied from facility $j \in J$.
- ▶ $y_j \in \{0, 1\}$: Decision to locate the health care facility in $j \in J$ (or not).

$$\min. \sum_j^J F_j y_j + \sum_i^I \sum_j^J C_{ij} D_i x_{ij} \quad (8.15)$$

$$\text{s.t.: } \sum_j x_{ij} = 1 \quad \forall i \in I \quad (8.16)$$

$$x_{ij} \leq y_j \quad \forall i \in I, j \in J \quad (8.17)$$

$$y_j \in \{0, 1\} \quad \forall i \in I, j \in J \quad (8.18)$$

$$x_{ij} \geq 0 \quad \forall i \in I, j \in J \quad (8.19)$$

The Stochastic Programming UFLP⁵ with uncertain **health requirements** and **travel time**⁶ [Contreras and Ortiz-Astorquiza, 2019]

$$\min. \sum_{j \in J} F_j y_j + Q(y) \quad (8.27)$$

$$\text{s.t.: } \sum_{j \in J} y_j \geq 1 \quad (8.28)$$

$$y_j \in \{0, 1\} \quad \forall i \in I \quad (8.29)$$

Where $Q(y) = \mathbb{E}_{\Xi}[Q(y, \xi)]$, and $Q(y, \xi)$ is solution of:

$$\min. \sum_{i \in I} \sum_{j \in J} C_{ij} D_i x_{ij} \quad (8.30)$$

$$\text{s.t.: } \sum_j x_{ij} = 1 \quad \forall i \in I \quad (8.31)$$

$$x_{ij} \leq y_j \quad \forall i \in I, j \in J \quad (8.32)$$

$$x_{ij} \geq 0 \quad \forall i \in I, j \in J \quad (8.33)$$

⁵Uncertainty can be measured probabilistically and included in a random vector Ξ .

⁶The model (8.30)–(8.33) is defined for every realization of demand and travel time ξ of Ξ

Extensive form of Deterministic Equivalent UFLP⁷

[Contreras and Ortiz-Astorquiza, 2019]

W : Finite set of scenarios ω in Ω .

⁷Non-anticipativity principle: each **first-stage** variables have **the same** value for all scenarios.

Extensive form of Deterministic Equivalent UFLP⁷

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W : Finite set of scenarios ω in Ω .

$$\min. \sum_{j \in J} F_j y_j + \sum_{\omega \in \Omega} \pi_{\omega} \left(\sum_{i \in I} \sum_{j \in J} C_{ij\omega} D_{i\omega} x_{ij\omega} \right) \quad (8.34)$$

$$\text{s.t.: } \sum_{j \in J} y_j \geq 1 \quad (8.28)$$

$$y_j \in \{0, 1\} \quad \forall i \in I \quad (8.29)$$

$$\sum_{j \in J} x_{ij\omega} = 1 \quad \forall i \in I, \omega \in \Omega \quad (8.35)$$

$$x_{ij\omega} \leq y_j \quad \forall i \in I, j \in J, \omega \in \Omega \quad (8.36)$$

$$x_{ij\omega} \geq 0 \quad \forall i \in I, j \in J, \omega \in \Omega \quad (8.37)$$

⁷Non-anticipativity principle: each **first-stage** variables have **the same** value for all scenarios.

The Deterministic Equivalent **Capacitated** Facility Location Problem (CFLP) ⁸[Contreras and Ortiz-Astorquiza, 2019]

I : The set of all patients' origins.

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⁸Constraint (8.32) is changed to constraint (8.38)

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- ▶ π_w : **Probability** associated to scenario ω
- ▶ $D_{i,\omega}$: Population requirement for health at origin $i \in I$ and scenario ω (pop).
- ▶ F_j : Setup cost for facility $j \in J$ (\$).
- ▶ G_j : Unit cost for installing facility $j \in J$ (\$).
- ▶ K_j : Unit Capacity of facility $j \in J$ (pop).
- ▶ $C_{ij\omega}$: Patient cost for meeting demand from $i \in I$ to $j \in J$ in ω (\$).
- ▶ PC_i : Penalty cost for Non-Satisfied Demand from origin $i \in I$ (\$).
- ▶ $R_{i\omega}$: Revenue from patient $i \in I$ in scenario ω (\$).

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- ▶ $y_j \in \{0, 1\}$: Decision to locate the health care facility in $j \in J$ (or not).
- ▶ $z_j \in \{0, 1\}$: Decision to locate **capacity** to health care facility in $j \in J$ (or not).
- ▶ $n_{i,\omega}$: **Non-satisfied** requirement for health at $i \in I$ and scenario ω (pop).

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The Deterministic Equivalent **Capacitated** Facility Location Problem (CFLP) ⁹[Contreras and Ortiz-Astorquiza, 2019]

$$\min. \sum_j^J F_j y_j + \sum_j^J G_j z_j + \sum_{\omega}^{\Omega} \pi_{\omega} \left(\sum_i^I P C_i n_{i\omega} + \sum_i^I \sum_j^J C_{ij\omega} (C_{ij\omega} - R_{i\omega}) D_{i\omega} x_{ij\omega} \right) \quad (8.39, 8.42, 8.44)$$

$$\text{s.t.: } \sum_{i \in I} x_{ij\omega} + n_{i\omega}/D_{i\omega} = 1 \quad \forall i \in I, \omega \in \Omega \quad (8.40)$$

$$\sum_{j \in J} x_{ij\omega} \leq 1 \quad \forall i \in I, \omega \in \Omega \quad (8.45)$$

$$D_{i\omega} x_{ij\omega} \leq K_j y_j \quad \forall i \in I, \omega \in \Omega \quad (8.36)$$

$$D_{i\omega} x_{ij\omega} \leq z_j \quad \forall i \in I, \omega \in \Omega \quad (8.46)$$

$$y_j \in \{0, 1\} \quad \forall i \in I, j \in J \quad (8.32)$$

$$x_{ij\omega} \geq 0 \quad \forall i \in I, j \in J, \omega \in \Omega \quad (8.33)$$

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References

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